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United States Patent Application Publication

20250277952

Kind Code

A1

Publication Date

September 04, 2025

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FIBER OPTIC TERMINALS, COMPENSATION RINGS, AND METHODS FOR ADJUSTING A SPLIT RATIO OF A FIBER OPTIC TERMINAL

Abstract

A fiber optic terminal having a variable ratio coupler that is adjusts power ratio splits using a rotating compensation ring along with related methods. In one embodiment, a compensation ring for a variable ratio fiber optic coupler includes a ring body and at least one arcuate slot within the ring body operable to receive at least one locking fastener. The compensation ring also includes a shaft opening within the ring body, where the at least one arcuate slot at least partially surrounds the shaft opening, the shaft opening defines an inner surface, and the inner surface includes a plurality of detent receiving features.

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Family ID: 96881227

Appl. No.: 19/056156

Filed: February 18, 2025

Related U.S. Application Data

us-provisional-application US 63559543 20240229

Publication Classification

Int. Cl.: G02B6/44 (20060101)

U.S. Cl.:

CPC G02B6/4472 (20130101);

Background/Summary

RELATED APPLICATIONS [0001] This application claims the benefit of priority of U.S. Provisional Application Ser. No. 63/559,543 filed Feb. 29, 2024, the content of which is relied upon and incorporated herein by reference in its entirety.

FIELD

[0002] The disclosure is directed to fiber optic terminals having variable ratio couplers and tools for changing the output power level of optical outputs along with fiber optic networks using the terminals.

BACKGROUND

[0003] Optical fiber is increasingly being used for a variety of applications, including but not limited to broadband voice, video, and data transmission. As bandwidth demands increase optical fiber is migrating deeper into communication networks such as in fiber to the premises applications such as FTTx, 5G and the like. As optical fiber extends deeper into communication networks there exists a need for building more complex and flexible fiber optic networks in a quick and easy manner.

[0004] Terminals such as multiports or closures were also developed for making one or more optical connections with hardened connectors such as the OptiTap® plug connector. Prior art multiports have an input cable or input port with a plurality of receptacles mounted through a wall of the housing for protecting an indoor connector inside the housing that makes an optical connection to the external hardened connector of the branch or drop cable.

[0005] Illustratively, FIG. 1 shows a conventional fiber optic multiport 1 having an input fiber optic cable 4 carrying one or more optical fibers to indoor-type connectors inside a housing 3. The multiport 1 receives the optical fibers into housing 3 and distributes the optical fibers to receptacles 7 for connection with a hardened connector. The receptacles 7 are separate assemblies attached through a wall of housing 3 of the multiport 1. The receptacles 7 allow mating with hardened connectors attached to drop or branching cables (not shown) such as drop cables for “fiber-to-the-home” applications. During use, optical signals pass through the branch cables, to and from the fiber optic cable 4 by way of the optical connections at the receptacles 7 of multiport 1. Fiber optic cable 4 may also be terminated with a fiber optic connector 5.

[0006] Multiport 1 allows quick and easy deployment by service providers for passive optical networks. Further, multiport 1 may use a coupler or splitter inside the multiport to allow a single input optical signal to be split into multiple output channels. By way of explanation, the input fiber optic cable may have a single optical fiber that is in optical communication with a 1:N splitter for outputting N output signals. However, the power level of the input optical channel is divided among the N output signals in a passive optical network (e.g., no active components are used in the passive portion of the optical network). By way of explanation, a 1:2 coupler may split the power from the single input optical fiber as 50% power for the first output optical signal and 50% power for the second output optical signal. Other couplers may have unequal splits in the power level as desired such as splitting the power from the single input optical fiber as 80% power for the first output optical signal and 20% power for the second output optical signal depending on the requirements for the fiber optic network. Furthermore, multiports may be daisy-chained together

for building more complicated fiber optic networks with further power level splits for the distribution of passive optical signals. By way of a simple explanation, an input optical signal from the central office may be able to accommodate a total split of 1:16 for the given input power level of the optical signal. An upstream multiport may have a 1:2 split with equal power levels for the two output fibers that each feed separate downstream multiports having a further 1:8 split with equal power levels, thus the single input fiber is split into 16 output signals each having an equal power level. Alternatively, a single multiport can incorporate a 1×2 splitter with unequal power split, with one output connected to a 1×N equal power splitter and the other connected to a downstream multiport.

[0007] However, conventional couplers or splitters have a fixed power level split for the output signals. This requires many individual couplers or splitters each having its own SKU, which increases both manufacturing and inventory costs. Moreover, fixed power level split does not readily allow for easy modification to the fiber optic network due to changed circumstances such as adding new customers or adapting the power levels needed for different loss budgets across the length of the passive optical network.

[0008] Consequently, there exists an unresolved need for terminals that provide quick and easily deployment for the fiber optic network in a flexible manner while also addressing concerns related to limited space, organization, or aesthetics.

SUMMARY

[0009] The disclosure is directed to fiber optic terminals (hereinafter “terminals”) and compensation rings for adjusting a split ratio of fiber optic terminals including variable ratio couplers. The compensation rings and the terminals with variable ratio couplers allow the power levels for the optical outputs from the variable ratio coupler to be changed as desired, thereby providing flexibility for the network operators to adapt or customize their network for their given needs.

[0010] In one embodiment, a compensation ring for a variable ratio fiber optic coupler includes a ring body and at least one arcuate slot within the ring body operable to receive at least one locking fastener. The compensation ring also includes a shaft opening within the ring body, where the at least one arcuate slot at least partially surrounds the shaft opening, the shaft opening defines an inner surface, and the inner surface includes a plurality of detent receiving features. The compensation ring may be used with variable ratio couplers such as variable ratio couplers disposed in a fiber optic device or fiber optic terminal.

[0011] In another embodiment, a fiber optic terminal includes a shell defining a cavity, and a plurality of ports including a control port having a port opening extending from an outer surface of the terminal into the cavity and defining a port passageway along a longitudinal axis. The fiber optic terminal also includes a variable ratio coupler disposed within the cavity including a housing, an optical input, a first optical output, a second optical output, a shaft within the control port, a detent radially extending from the shaft, and a compensation ring disposed around the shaft. The compensation ring includes a ring body, at least one arcuate slot within the ring body operable to receive at least one locking fastener, and a shaft opening within the ring body. The compensation ring is secured to the housing by one or more locking fasteners. The at least one arcuate slot at least partially surrounds the shaft opening. The shaft opening defines an inner surface that includes a plurality of detent receiving features. Rotation of the shaft changes an output power level between the first optical output and the second optical output at a coupling region, and the detent is operable to be positioned in a selected detent feature of the plurality of detent features.

[0012] In another embodiment, a method of changing a split ratio of a fiber optic terminal includes inserting an axle of a tool into a control port passageway of a control port of a plurality of ports of the fiber optic terminal to engage an end of the axle with a shaft of a variable ratio coupler within a cavity of the fiber optic terminal, and rotating the axle of the tool to rotate the shaft of the variable ratio coupler to a desired split ratio set point of a plurality of split ratio set points. The desired split

ratio set point corresponds to a desired output power level between a first optical output and a second optical output of the variable ratio coupler, and each split ratio set point is defined by a detent receiving feature of a plurality of detent receiving features of a compensation ring surrounding the shaft.

[0013] In another embodiment, a variable ratio coupler for changing a split ratio of optical power for two optical fibers includes a coupler having a coupling region operable to change the split ratio of the optical power for the two optical fibers. The variable ratio coupler also includes a housing. The variable ratio coupler further includes a shaft disposed within the housing that includes an end, a radially extending detent, an end face at the end configured to receive a tool, where rotation of the shaft by the tool changes an output power level between the first optical output and the second optical output at a coupling region. The variable ratio coupler further includes a compensation ring disposed around the shaft, the compensation ring having a ring body and at least one arcuate slot within the ring body. The variable ratio coupler also includes at least one locking fastener disposed within the at least one arcuate slot that secures the compensation ring to the housing and a shaft opening within the ring body, where the at least one arcuate slot is at least partially surrounds the shaft opening, the shaft opening defines an inner surface, and the inner surface includes a plurality of detent receiving features operable to receive the detent of the shaft. The variable ratio coupler also includes a cam disposed on the shaft, where the cam has a shape that changes an amount of bend on the coupling region for a desired split ratio of optical power for the two optical fibers. The plurality of detent receiving features corresponds to a plurality of split ratios of optical power.

[0014] In another embodiment, a method of assembling a variable ratio coupler includes positioning a compensation ring over a shaft such that a detent is positioned within a detent receiving feature corresponding to a first split ratio set point, inputting an optical signal into an optical input of the variable ratio coupler, measuring an optical power at a first optical output and a second optical output, turning the shaft until a split ratio of optical power between the first optical output and the second optical output is at the first split ratio optical power set point, and securing the compensation ring to a housing of the variable ratio coupler.

[0015] In another embodiment, a method of assembling a variable ratio coupler includes positioning a compensation ring over a shaft such that a detent is positioned within a detent receiving feature corresponding to a first split ratio set point, inputting an optical signal into an optical input of the variable ratio coupler, measuring an optical power at a first optical output and a second optical output, turning the shaft until a split ratio of optical power between the first optical output and the second optical output is at the first split ratio optical power set point, and recording the position of the shaft. The method also includes sequentially turning the shaft until the split ratio of optical power is at a plurality of split ratio set points and recording the position of the shaft at each individual split ratio set point of the plurality of split ratio set points. The method further includes removing the compensation ring from the shaft, fabricating a plurality of detent receiving features on the compensation ring according to the recorded positions of the shaft at the plurality of split ratio set points, and positioning the compensation ring over the shaft such that the detent is positioned within the dent receiving feature corresponding to the first split ratio set point. The method also includes inputting the optical signal into the optical input of the variable ratio coupler, measuring the optical power at the first optical output and the second optical output, turning the shaft until the split ratio of optical power between the first optical output and the second optical output is at the first split ratio optical power set point, and securing the compensation ring to a housing of the variable ratio coupler.

[0016] Additional features and advantages will be set forth in the detailed description which follows, and in part will be readily apparent to those skilled in the art from that description or recognized by practicing the same as described herein, including the detailed description that follows, the claims, as well as the appended drawings.

[0017] It is to be understood that both the foregoing general description and the following detailed

description present embodiments that are intended to provide an overview or framework for understanding the nature and character of the claims. The accompanying drawings are included to provide a further understanding of the disclosure, and are incorporated into and constitute a part of this specification. The drawings illustrate various embodiments and together with the description serve to explain the principles and operation.

Description

BRIEF DESCRIPTION OF THE FIGURES

[0018] FIG. 1 is a prior art multiport;

[0019] FIG. 2 is a schematic representation of a fiber optic network having terminals with a variable ratio coupler;

[0020] FIG. 3 is a schematic representation of the variable ratio coupler depicted in the terminals of FIG. 2;

[0021] FIG. 4 is a perspective view of an example fiber optic terminal comprising a variable ratio coupler controllable by a tool and with a plurality of output connection ports;

[0022] FIG. 5 is a partially exploded view of another explanatory terminal showing further details of a specific construction for terminals;

[0023] FIG. 6 is a transparent top view of another explanatory terminal showing further details of a specific construction for terminals;

[0024] FIG. 7 is a perspective view of an explanatory tool for changing an output power level between a first optical output and a second optical output;

[0025] FIG. 8 is a perspective view of an explanatory variable ratio coupler for receiving a tool to change an output power level between a first optical output and a second optical output;

[0026] FIG. 9 is a cross-sectional view of an explanatory variable ratio coupler for receiving a tool to change an output power level between a first optical output and a second optical output;

[0027] FIG. 10 is a front elevation view of an explanatory variable ratio coupler having a shaft and a compensation ring;

[0028] FIG. 11 is a front elevation view of another explanatory compensation ring;

[0029] FIG. 12 is a graph illustrating coupling ratio as a function of wavelength for a plurality of deflection values of a variable ratio coupler; and

[0030] FIG. 13 is a front elevation view of an explanatory cam profile of a shaft of a variable ratio coupler.

DETAILED DESCRIPTION

[0031] References will now be made in detail to the embodiments of the disclosure, examples of which are illustrated in the accompanying drawings. Whenever possible, like reference numbers will be used to refer to like components or parts.

[0032] The concepts disclosed are related to fiber optic networks, tools and fiber optic terminals having at least one variable ratio coupler with shaft for actuation by a tool for changing an output power level between a first optical output and a second optical output for a passive optical network. As used herein, “variable ratio coupler” means that the output power level may be adjusted to many different power level splits (i.e., “split ratios”) across the spectrum of output power levels so that the power level split may be tuned or changed by the user as desired in a passive operation that does not require power to be supplied to the module for its operation, and does not mean the power level may only be changed to two discrete power level splits. Consequently, the fiber optic terminals (hereinafter “terminals”) comprising the variable ratio coupler(s) (hereinafter “VRC(s)”) are well-suited for passive optical networks such as in the outside plant environment such as downstream from a central office location or the like, but other applications are possible.

[0033] In addition to the passive operation and providing a wide range of possible output power

split levels, the concepts using the VRC disclosed provide a stable performance across varying conditions. Further, the terminals and networks using the VRC have a low polarization dependent loss (PDL). In other words, the polarization state of the optical signal does not adversely impact the performance of the terminals or networks. Thus, the polarization state of the optical signal is not a factor for performance or operation. By way of example, the PDL loss is about 0.5 dB or less, and may even be as low as 0.3 dB or less or 0.2 dB or less for any polarization state of the input optical signal.

[0034] Still further, terminals and networks using the VRC have a wide wavelength range for suitable performance. By way of example, the terminals and networks using VRC comprise a similar performance from about 1260 nm to about 1625 nm. Generally speaking, the terminals disclosed and explained in the exemplary embodiments are multiports, but the concepts disclosed may be used with any suitable terminal such as closures, network interface devices, wireless radios or the like having at least one variable ratio coupler with a control for changing an output power level.

[0035] The concepts disclosed advantageously provide flexibility for the network operators and also reduce manufacturing complexity and inventory concerns for manufacturers of the terminals along with network operators since the need to manufacture and stock a multitude of terminals having different fixed power split levels is not necessary. In other words, the terminals and fiber optic networks disclosed may be adjusted to have the desired power level splits at any point during their lifetime, thereby providing flexibility and adaptability to alter the fiber optic network based on moves, adds or changes to the fiber optic network. The concepts may be used with any suitable terminals and may be especially advantageous with terminals having compact form-factors. The concepts are scalable to any suitable count of input or outputs on a terminal in a variety of arrangements or constructions for building fiber optic networks.

[0036] For instance, the concepts disclosed herein are suitable for fiber optic networks such as for Fiber-to-the-Home and 5G applications and are equally applicable to other optical applications as well including indoor, industrial, wireless, or other suitable applications. The concepts disclosed herein are especially advantageous for asymmetric split fiber optic networks (e.g., fiber optic networks having one VRC with an unequal output power level split). Additionally, the concepts disclosed may be used with terminals having any suitable footprint or construction.

[0037] Although VRC fiber optic terminals are desirable, providing the mechanisms to adjust the split ratio of the VRC in each fiber optic terminal may undesirably increase the cost of the fiber optic terminal. For example, more parts need to be fabricated and incorporated into the fiber optic terminal, thus increasing its cost of manufacture. Embodiments of the present disclosure reduce the overall cost to manufacture a fiber optic terminal having a VRC by including at least some of the mechanics for adjusting the split ratio of VRCs into a separate tool. Thus, a single tool may be used to adjust the split ratio of many fiber optic terminals. For example, the set-point display may be incorporated into the tool itself, as well as the mechanical means to adjust a coupling region of the VRC.

[0038] The embodiments described herein also enable a manufacturer of telecommunications equipment to replace multiple SKUs, each representing a terminal with a non-variable coupler with a fixed tap ratio, with a single SKU number for the fiber optic terminal having a VRC. This further reduces the costs to manufacture, purchase and manage fiber optic terminals incorporating couplers.

[0039] Various designs, constructions, or features for fiber optic networks, tools and terminals are disclosed in more detail as discussed herein and may be modified or varied as desired.

[0040] FIG. 2 shows a schematic view of an explanatory fiber optic network **11** such as for a passive fiber-to-the-home (FTTH) network comprising a first terminal **200** having a VRC; however, the concepts may be used with other networks such as a PON, FTTx or 5G networks. As depicted, a first optical link **10a** (e.g., a first fiber optic cable) is an input optical link connected to a

central office CO at a first end and a second end is in optical communication with the optical input OI of the first terminal **200**. A first end of a second optical link **10b** (e.g., a second optical cable) is an output optical link of terminal **200** and is in optical communication with the first optical output (OT1) of the terminal **200** as depicted. A second end of the second optical link **10b** is in optical communication with the optical input OI of a second terminal **200'**. A first end of a third optical link **10c** (e.g., a third optical cable) is an output optical link of terminal **200'** and is in optical communication with the first optical output (OT1) of the terminal **200'** as depicted. A second end of the third optical link **10c** feeds into the optical input OI of a third terminal **200''**. A first end of a fourth optical link **10d** (e.g., a fourth optical cable) is an output optical link of terminal **200''**. The fiber optic network **11** splits the power level launched from the CO at the respective terminals **200**, **200'** and **200''** for the distribution of optical signals to the fiber optic network **11**.

[0041] Terminals **200**, **200'** and **200''** are schematically depicted in FIG. 2 each of which comprises a shell **210** having a cavity **216** with a portion of the respective VRCs being disposed within the respective cavities **216**. The terminals **200**, **200'** and **200''** also comprise at least one input connection port **236**, and a plurality of ports **260**. The VRCs each also comprise the optical input (OI), the first optical output (OT1), the second optical output (OT2) and a control (CTL) for changing an output power level between the first optical output (OT1) and the second optical output (OT2) as depicted. The input connection port **236** may comprise a port opening **238** extending from an outer surface (**234**) of the terminal **200** into the cavity **216** and defines a port passageway **233** along a longitudinal axis. In this embodiment, terminals **200**, **200'** and **200''** of fiber optic network **11** comprise the same configuration as depicted; however, the VRCs are adjusted with different output power level split between the respective first optical output (OT1) and second optical output (OT2) using the control (CTL). As described in more detail below, the control (CTL) may be adjusted using a tool, and further the control (CTL) may be provided within one of the optical ports of the terminal **200**, **200'**, **200''**. The output power level split for the VRC may be asymmetric or not depending on the desired output power levels.

[0042] By way of explanation, fiber optic network **11** distributes the signal from the second optical output (OT2) from the respective VRCs to each local neighborhood where the bandwidth of the optical output is shared by multiple subscribers. For instance, terminal **200** may have its VRC adjusted to a 90/10 split of the power received from the central office (CO) (minus losses) with 90% of the input power being directed to the first optical output (OT1) and 10% of the input power being directed to the second optical output (OT2) for the distribution of optical signals to its local neighborhood. Thereafter, terminal **200'** receives 90% of the power transmitted to its optical input (OI) (minus losses such as connector losses, transmission losses, etc.), and may have its VRC adjusted to a 75/25 split of the power received at its optical input (OI) with 75% of the input power to terminal **200'** being directed to its first optical output (OT1) and 25% of the input power being directed to its second optical output (OT2) for the distribution of optical signals to its local neighborhood. Terminal **200''** that receives 75% of the power from the optical output (OT1) of terminal **200'** at the optical input (OI) may have its VRC adjusted to a 50/50 split of the power received with 50% of the input power being directed to the first optical input (OT1) and 50% of the input power being directed to the second optical output (OT2) for the distribution of optical signals to its local neighborhood. This representative fiber optic network **11** allows the desired power levels to be transmitted to the local neighborhoods, while transmitting the remaining power downstream in the fiber optic network **11** as desired. Moreover, the output power level split ratios within the terminals **200**, **200'** and **200''** may be easily and quickly adjusted by the network operator as needed for moves, adds or changes in the fiber optic network **11** as desired, thereby providing flexibility and adaptability that is lacking in conventional fiber optic networks.

[0043] FIG. 3 is a schematic representation of the VRC depicted in terminals **200**, **200'** and **200''**. As depicted, VRC comprises an optical input (OI) that has its output power level split between the first optical output (OT1) and the second optical output (OT2) at a coupling region (CR), with the

control (CTL) for changing the output power level between the first and second outputs (OT1,OT2). The coupler may be a planar lightwave circuit (PLC) or multicladd coupler (MC) as known in the art, but other suitable structures may be used. The optical input (OI) and the optical outputs (OT1, OT2) are optical waveguides such as optical fibers that may be in optical communication with the respective input and outputs of the planar lightwave circuit or other type of device. Control (CTL) may be actuated (e.g., by use of a tool) for changing the output power level between the first optical output (OT1) and the second optical output (OT2) by any suitable means at the coupling region (CR).

[0044] The coupling region (CR) is the region where a portion of the first optical waveguide of the first optical output (OT1) and a portion of the second optical waveguide of the second optical output (OT2) are in optical (e.g., intimate) contact for allowing the changing of the output power level of the optical signals transmitted by the first optical output (OT1) and the second optical output (OT2). More specifically, the control (CTL) is configured for individually moving a portion of the first optical waveguide of the first optical output (OT1) (or alternatively moving a portion of the second optical waveguide of the second optical output (OT2)) at the coupling region (CR) as represented by the horizontal line with the arrows on the ends. The individual movement of the first or second optical waveguide with the control (CTL) may bend, deflect or change the geometry between the portion of the first optical waveguide of the first optical output (OT1) and the portion of the second optical waveguide of the second optical output (OT2) at the coupling region (CR) (i.e., an amount of deflection of the coupling region) for changing the output power level of the optical signals transmitted by the first optical output (OT1) and the second optical output (OT2). In further embodiments, the portion of the first optical waveguide and the portion of the second optical waveguide are fused together at the coupling region (CR) and the coupling ratio may be changed, for example, by bending the fused region. Other constructions are possible for the coupling region (CR) for changing the output power level of the optical signals transmitted by the first optical output (OT1) and the second optical output (OT2). For instance, other embodiments may change the index of refraction of the materials in or around the coupling region (CR). It should be understood that embodiments described herein are not limited by the method of changing the coupling ratio, such as those described in U.S. Pat. No. 7,627,214, for example.

[0045] FIG. 4 depicts an explanatory terminal **200** that comprises at least one input connection port **236** and a pass-through connection port **260PT**. Input connection port **236** and pass-through output connection ports **260PT** are suitable for receiving respective external fiber optic connectors **100** of the optical link **10a**. In this construction, the terminal **200** comprises an optical link **10a** configured as a fiber optic cable that is secured to the input connection port **236** as a tether cable and optically connected to the optical input (OI) of the VRC. In other words, the fiber optic cable is not intended to be removable from the input connection port **236**. The other end of the tether cable may be terminated with a suitable fiber optic for optical connectivity to the fiber optic network.

[0046] On the other hand, the pass-through connection port **260PT** of terminal **200** of FIG. 4 is in optical communication with the first optical output (OT1) of the VRC. Terminal **200** of FIG. 4 also comprises a second coupler (C2) in optical communication with the second optical output (OT2) of the VRC such as schematically depicted in FIG. 2. The second optical coupler (C2) comprises a plurality of second coupler outputs (C201-C20x), and the second coupler outputs (C201-C20x) are in optical communication with a plurality of optical connection ports **260**. More specifically, the second coupler outputs may comprise optical fibers extending from a PLC that are optically connected or terminated with respective fiber optic connectors **252** disposed within the cavity **216** of the terminal and are aligned with the respective port **260** for optical connection with the terminal **200**. Terminal **200** of FIG. 4 comprises six output connection ports **260**, but terminals **200** may use any suitable number of output connection ports as desired. The output connection ports **260** may be optically connected to drop cables having a suitable connector for routing the optical signals toward the subscribers.

[0047] In the embodiment of FIG. 4, the control port **260CTL** is provided as an optical port opening such that it is within an array of output connection ports **260**. Such a placement for the **260CTL** may be advantageous because special shells do not need to be molded specifically for terminals including a VRC. Thus, one SKU number may be provided for shells intended for variable split ratio terminals and for shells intended for fixed split ratio terminals. As described in more detail below, the split ratio of the terminal **200** is adjusted by inserting a tool into the control port **260CTL**. Another advantage of the embodiment shown by FIG. 4 is that the same plugs can be used for the control port **260CTL** as the output connection ports **260**.

[0048] In further explanation the terminal **200** of FIGS. 4-6 comprises a shell **210** with a cavity **216** along with a securing feature **310** comprising a securing member **310M** associated with the port passageway **233** (FIG. 5). The input connection port **236**, and pass-through connection ports **260PT** each comprise a port opening extending from an outer surface of the terminal **200** into the cavity **216** of the terminal **200** and each port respectively defines a port passageway along a longitudinal axis. Each port **236**, **260PT** has a respective securing member **310M** associated with the port. Each securing member **310M** comprises a bore **310B** suitable for receiving and securing a portion of the housing **20** of the fiber optic connector of the respective optical link such as depicted with the input optical link **10a** inserted into the input connection port **236**. Likewise, the output connection ports **260** where used may have a similar construction as described for the input connection port **236** and pass-through connection ports **260PT**. Terminals **200** may also advantageously use the securing members **310M** for releasably connecting the external fiber optic connectors **100** of the optical links in the respective connection ports using an actuator **310A** of securing feature **310**.

[0049] As stated above, the split ratio provided by a variable ratio coupler within the terminal is adjusted by a tool that is inserted into the cavity **216** of the shell **210**. FIG. 5 illustrates a VRC **450** within the cavity **216** according to one embodiment. The VRC **450** is accessible by a control port **260CTL** that is configured as an output port and is in an array of output ports **260**. The control port **260CTL** is defined by a passageway extending from an edge of the shell **210** into the cavity. As described in more detail below, the control port **260CTL** is operable to receive a tool that engages a shaft **452**. FIG. 4 illustrates an example tool **400** positioned in a control port **260CTL**. Rotation of the tool **400** causes rotation of the shaft **452**, which in turn causes the coupling region within the VRC **450** to bend. The bending of the coupling region changes the output power level of optical outputs OT1 and OT2 and thus changes the split ratio between optical outputs OT1 and OT2.

[0050] FIG. 4 illustrates a tool **400** inserted into a control port **260CTL**. Particularly, the tool **400** comprises an axle **402** and an end piece, which is configured as a knob **401** in the illustrated embodiment. The user inserts the axle **402** into the passageway of the control port **260CTL** until it engages the VRC. The user then applies a rotational force to the knob **401**, which rotates the axle **402** and a shaft of the VRC to change the split ratio as described in more detail below.

[0051] FIG. 6 illustrates a partially transparent top view of a terminal **200** showing internal components of the terminal **200**. Rather than a rear connector **252** at port #12, the terminal includes a VRC **450** at port #12. When inserted into the control port **260CTL** (i.e., port **12**), the tool **400** engages the VRC **450** and enables the user to change the split ratio of the terminal **200** accordingly.

[0052] FIG. 7 illustrates a non-limiting example of a tool **400** for adjusting a split ratio of a terminal **200**. The example tool **400** includes an axle **402** for insertion into a port (i.e., a control port **260CTL**) of a fiber optic terminal. An end **407** of the axle **402** is rotationally asymmetric such that it is insertable into the VRC in only one rotational orientation. The phrase “rotationally asymmetric” means a component has only one rotational orientation with respect to another component. In the embodiment of FIG. 7, the end **407** includes a peak edge **409** such that the peak edge may only be inserted into a similarly shaped notch within the VRC.

[0053] The tool **400** further includes a terminal engagement body **403** disposed about the axle **402**. The axle **402** is free to at least partially rotate about the terminal engagement body **403**. The terminal engagement body **403** includes a terminal engagement feature **404** for engaging a

corresponding alignment feature of the terminal **200**. In the illustrated embodiment, the terminal engagement feature **404** is configured as a notch that is operable to engage an alignment feature configured as a protrusion, such as a ledge or a post. FIG. 5 illustrates an example alignment feature **213** configured as a ridge-like protrusion within the passageway defined by the control port **260CTL**. Thus, the terminal engagement feature **404** and the alignment feature **213** cooperate to index the terminal engagement body **403** to a known position with respect to the terminal **200** and thus the VRC. It should be understood that other configurations for the terminal engagement feature **404** and the alignment feature **213** are possible.

[0054] Referring once again to FIG. 8 the terminal engagement feature **404** further comprises a set-point portion **405** that includes a set-point indicator **406**. The set-point indicator **406** is referenced to the terminal engagement feature **404**. As a non-limiting example, the set-point indicator **406** is aligned with the terminal engagement feature **404** along a longitudinal axis of the tool **400**. The illustrated set-point indicator **406** is a ridge formed within the set-point portion **405**. However, in other embodiments the set-point indicator **406** may be a marking, for example (e.g., an arrow, a line, a circle, etc.).

[0055] The tool **400** also includes an end piece configured as a knob **401** that is rigidly coupled to the axle **402**. Rigidly coupled means that the knob **401** is not free to rotate with respect to the axle **402**. The knob **401** has the function of providing a means for the user to rotate the tool **400** as well as providing a read-out of the current set-point of the terminal **200**. Although FIG. 8 illustrates the end piece as a knob **401**, embodiments are not limited thereto.

[0056] The knob **401** includes a plurality of set-point markers **408** that indicate which set-point (i.e., what split ratio) the terminal **200** is presently set at. In the illustrated embodiment, the set-point markers **408** comprise lines and associated numbers indicating the set-point. Embodiments are not limited to any particular marker for the set-point markers. For example, in another embodiment, the numbers may be replaced by the actual split ratio of the particular set point (e.g., **90/10**, **75/25**, and the like).

[0057] The terminal engagement body **403** is also free to rotate with respect to the knob **401** because the knob **401** is rigidly coupled to the axle. The knob **401** may be rotatably coupled to the terminal engagement body **403** by any manner. In the illustrated embodiment, the knob **401** includes an arcuate slot **411** extending from a face of the knob **401** into a passageway defined by the knob **401**. An attachment feature **410** extends from an end face of the terminal engagement body **403** that is disposed within the passageway of the knob **401**. The attachment feature **410** extends through the slot and prevents the knob **401** from being pulled away from the terminal engagement body **403**. As a non-limiting example, a small screw may be applied to the attachment feature **410** after the attachment feature is positioned through the slot **411** to prevent the knob **401** from being separated from the terminal engagement body **403**. The attachment feature **410** and the slot **411** prevent longitudinal movement of the knob **401** while allowing rotational movement of the knob **401** and axle **402** with respect to the terminal engagement body **403**. Attachment feature **410** also provides limit on the rotational amount of knob **401** and said feature is not typically separable in normal use.

[0058] Upon insertion of the tool **400** into a control port **260CTL**, the terminal engagement feature **404** engages the alignment feature **213** (FIG. 5) associated with the terminal **200**. This orients the terminal engagement feature **404** to a proper and known position with respect to the terminal **200**. Thus, the set-point indicator **206** is in a measurement position with respect to the terminal.

[0059] As the user continues to insert the tool **400** into the control port **260CTL**, the end **407** of the axle **402** reaches a shaft **452** (FIG. 8) of the VRC. Because the end **407** of the axle **402** is rotationally asymmetric, it can only be inserted into the shaft **452** of the VRC in one rotational orientation. The user may need to rotate the knob **401** to find the proper orientation of the axle **402** with respect to the shaft **452**. Once the axle **402** is properly seated in the shaft **452**, the user may turn the knob **401**, which also turns the axle **402** and the shaft **452**, to align the desired set-point

marker of the plurality of set-point markers **408** to the set-point indicator **406**. It should be understood that all possible embodiments of the shaft **407** are not shown but the concept is to discourage tampering by using a shape which though available from vendors in the telecom space is not commonly available to members of the public.

[0060] Referring now to FIG. **8**, a non-limiting example of a VRC **450** is illustrated. The VRC **450** includes a housing **451** that defines a cavity in which the internal components are disposed. It should be understood that embodiments are not limited to the particular VRC **450** illustrated by FIG. **8**, and that VRCs of the present disclosure may include different components or differently arranged components.

[0061] The VRC **450** includes a shaft **452** extending into the cavity defined by the housing **451**. The shaft **452** includes an end face **454** that is accessible through an opening of the housing **451**. The end face **454** is on a set-point portion **453** of the shaft **452**. The shaft **452** further includes a cam **458**, as described in more detail below. The shaft **452** further includes a rotationally asymmetric feature so that the axle **402** may be coupled to the shaft **452** in only one rotational position. In the illustrated embodiment, the rotationally asymmetric feature is a single hole **469** or opening that is rotationally asymmetric. The hole **469** is circular in shape but also includes a peak edge notch that is configured to receive a peak edge of the axle **402** in only one orientation. Embodiments are not limited to any particular rotationally asymmetric feature

[0062] The VRC **450** further includes a flexure **460** that is held down on one side by a bias member **459** configured as a spring. The other side of the flexure **460** contacts a surface of the cam **458**. An end **462** of the flexure **460** that is opposite from the end **461** that contacts the shaft **452** is rigidly coupled to the housing **451**. An amount of bend on the coupling region (CR) imparted by the flexure **460** determines the amount of optical power that is provided to a first output OT1 and to a second output OT2 and thus determines the split ratio. The cam **458** is designed such that different positions on the surface of the cam **458** provide differing amounts of bend on the coupling region (CR).

[0063] During use, an operator will use a tool or otherwise turn the shaft **452** to a position that corresponds with a desired set point. The VRC **450** is configured to snap into discrete set points as the shaft **452** is turned. Haptic feedback is provided to the operator when a set point is reached. In embodiments of the present disclosure, the set points and corresponding haptic feedback are provided by a compensation ring **455** that is disposed around a set point portion **453** of the shaft **452**.

[0064] FIGS. **9** and **10** illustrate a VRC **450** including a compensation ring **455**. FIG. **9** is a cross-sectional view of the VRC **450** while FIG. **10** is a front close-up view of the compensation ring and VRC **450**. The compensation ring **455** is disposed around the front end **454** of the shaft **452** such that the shaft **452** is free to rotate with respect to compensation ring **455**. The shaft **452** further includes a radially extending detent **470**. As a non-limiting example, the detent **470** may be configured as a ball detent. The detent **470** may be biased by a bias member **471**, for example. The bias member **471** may enable the detent **470** to be pushed in a direction toward the surface of the shaft **452** by an external force. The detent **470** is operable to be positioned within one of a plurality of detent receiving features, which correspond to individual split ratio set points for the VRC. FIG. **9** illustrates the detent **470** disposed within a first detent receiving feature **475-1**, which corresponds with a first split ratio set point.

[0065] Referring particularly to FIG. **10**, the compensation ring **455** is circular in shape and operable to be positioned within a recess of a housing **464** of the VRC **450**. The compensation ring **455** has a central shaft opening **476** that is operable to receive the front end **454** of the shaft **452**. The shaft **452** is free to rotate within the shaft opening **476**.

[0066] The compensation ring **455** further includes at least one arcuate slot that surrounds the shaft opening **476**, which in the illustrated embodiment includes a pair of arcuate slots including a first arcuate slot **473A** and a second arcuate slot **473B**, each of which oppose one another. In other

embodiments, there may be only a single arcuate slot. As described in more detail below, the arcuate slots enable rotation of the compensation ring **455** with respect to the housing **464** for calibration of the compensation ring **455** during a VRC assembly process.

[0067] The shaft opening **476** defines an inner surface **477** that faces the exterior surface of the shaft **452**. A plurality of detent receiving features **475** are located within the inner surface **477**. The detent receiving features **475** are configured as recesses into which the detent **470** of the shaft **452** may be positioned. The example compensation ring **455** includes a first detent receiving feature **475-1**, a second detent receiving feature **475-2**, a third detent receiving feature **475-3**, a fourth detent receiving feature **475-4**, a fifth detent receiving feature **475-5**, a sixth detent receiving feature **475-6** and a seventh detent receiving feature **475-7** (collectively referred to herein as detent receiving features **475**). It should be understood that embodiments are not limited by the number of detent receiving features **475**. The detent receiving features **475** may be blind bores as illustrated by FIG. **10**, or through holes.

[0068] Each detent receiving feature **475** corresponds to a particular set-point marker on the knob **401** and thus a particular split ratio set point (e.g., **90/10**, **75/25** etc.). Thus, the plurality of detent receiving features **475** correspond to a plurality of split ratio set points of the VRC **450**. As stated above, the VRC **450** further includes a detent that is biased toward the inner surface **477** by a bias member **471**, such as a coil spring, for example. In other embodiments no bias member is provided. As the shaft **452** is turned by the tool **400**, the detent **470** is seated and then exits the detent receiving features **475**. The detent **470** and the detent receiving features **475** provide haptic feedback to the operator so that the operator knows when the detent **470** is properly seated in a detent receiving feature **475** (e.g., a click or a vibration may be felt by the user through the tool **400** when the detent **470** enters a detent receiving feature **475**). Further, it will require more force by the user to turn the knob **401** to cause the detent **470** to exit the detent receiving feature **475** as compared to when the detent **470** is not present within a detent receiving feature **475**. In other words, a resistance to rotate the tool **400** is increased when the detent **470** is within a detent receiving feature **475** to move the detent **470** out of the detent receiving feature **475**. Additionally, the placement of the detent **470** within a detent receiving feature **475** ensures that shaft **452** is locked at the desired rotational position and split ratio set-point.

[0069] The compensation ring **455** is fabricated such that each detent receiving feature **475** corresponds with an individual split ratio set point as determined by the rotational position of the shaft **452**. The spacing between individual detent receiving features may be determined by a calibration process. Once the proper locations of the detent receiving features **475** are known, the compensation rings **455** may be fabricated, such as by injection molding, for example. The compensation ring **455** shown in FIG. **10** is universal and mass produced. As described in detail below with respect to FIG. **11**, in other embodiments the positions of the detent receiving features **475** may be determined for each individual VRC, thereby providing a specialized compensation ring **455'** for each individual VRC that is more precise than a universal compensation ring **455**.

[0070] The compensation ring **455** should be rotationally positioned within the VRC **450** at a known reference position such that the plurality of detent receiving features correspond with the proper plurality of split ratio set points. During an assembly process, a compensation ring **455** is disposed on a shaft **452** of the VRC **450** such that the shaft **452** is positioned through the shaft opening **476** of the compensation ring **455**, and the detent **470** is positioned within the first detent receiving feature **475-1**. One or more fasteners are then positioned within the one or more arcuate slots, such as a first fastener **474A** within a first arcuate slot **473A** and a second fastener **474B** within a second arcuate slot **474B**. The front face of the housing **464** includes threaded holes (not shown) to receive the one or more fasteners. The one or more fasteners are loosely threaded to the one or more threaded holes but not fully tightened so that the compensation ring **455** rotates with the shaft **452**.

[0071] An optical signal (e.g., laser light) is inputted into the optical input OI of the VRC **450**, and

the optical power at both the first optical output OT1 and the second optical output OT2 are measured. The shaft **452** is turned until the optical power of the first optical output OT1 and the second optical output OT2 provide a split ratio that is at the first split ratio set point (e.g., **90/10**). Next, the compensation ring **455** is secured to the housing **464** by tightening the one or more fasteners, such as the first fastener **474A** and the second fastener **474B** of FIG. **10**. Now the locations of the detent receiving features **475** are properly indexed to the positions of the shaft **452** that produce split ratios corresponding to the desired split ratio set points. The operator may then use the tool to rotate the shaft **452** such that the detent **470** is seated in the detent receiving feature **475** associated with the desired split ratio set point.

[0072] FIG. **11** illustrates an example compensation ring **455'** that is calibrated and fabricated for use with a specific VRC **450**. A specifically calibrated compensation **455'** may provide more precise split ratios than the mass-produced compensation ring **455** of FIG. **10**. A drill bit **500** or other material removing tool is used to fabricate through holes within the compensation ring **455'** body that correspond with the precision locations of split ratio set points for the intended VRC **450**. These holes define the plurality of detent receiving features, such as a first detent receiving feature **475-1'**, a second detent receiving feature **475-2'**, a third detent receiving feature **475-3'**, a fourth detent receiving feature **475-4'**, a fifth detent receiving feature **475-5'**, a sixth detent receiving feature **475-6'** and a seventh detent receiving feature **475-7'** (collectively referred to herein as detent receiving features **475'**). It should be understood that any number of detent receiving features **475'** may be utilized depending on the number of desired split ratio set points.

[0073] To determine the proper location for the detent receiving features, an optical signal (e.g., laser light) is inputted into the optical input OI of an individual VRC **450**, and the optical power at both the first optical output OT1 and the second optical output OT2 are measured. The shaft **452** is turned until the optical power of the first optical output OT1 and the second optical output OT2 provide a split ratio that is at the first split ratio set point (e.g., **90/10**). The position of the shaft **452** that provides the first split ratio set point is then recorded. The shaft **452** is then turned until the optical power of the first optical output OT1 and the second optical output OT2 provide a split ratio that is at the second split ratio set point (e.g., **80/20**). The position of the shaft **452** that provides the second split ratio set point is then recorded. This process is continued until the positions of the shaft **452** corresponding to all of the desired split ratio set points are determined and record.

[0074] Next, the detent receiving features **475'** corresponding to the recorded positions of the individual VRC **450** are fabricated in the compensation ring **455'**. For example, a drill bit **500** or other tool is used to fabricate through holes at the recorded positions that define the detent receiving features **475'**. In another embodiment, a compensation ring **455'** having detent receiving features **475'** at the recorded positions is fabricated by three-dimensional printing. Other methods for fabricating the compensation ring **455'** may be utilized. The detent receiving features **475'** may be through holes or blind bores, for example.

[0075] The specialized compensation ring **455'** may then be assembled to the VRC **450** by the same process as described about with respect to FIG. **10**.

[0076] FIG. **12** illustrates plots of the coupling ratio as a function of wavelength (nm) for a plurality of deflection position values of the coupling region CR. Each detent receiving feature **475** may correspond with a desired coupling ratio set point.

[0077] As stated above, the shaft **452** includes a cam **458** having a shape profile to impart various deflection values on the flexure **460** and thus the coupling region CR of the waveguide of the VRC **450**. FIG. **13** illustrates an example shape profile of a cam **458**. Line **500** represents the set point corresponding to the pre-molded reference position of the first detent receiving feature **475-1**, line **501** represents the set point corresponding to the second detent receiving feature **475-2**, line **502** represents the set point corresponding to the third detent receiving feature **475-3**, and line **503** represents the set point corresponding to the fourth detent receiving feature **475-4**.

[0078] Due to the nature of the cam profile, the 0 mm deflection position (i.e., corresponding with

the first detent receiving feature 475-1) may be achieved with a range of rotational values, whereas the preset coupling ratios only exist at a discrete rotary position, thus the location of the “0 deflection” position is forgiving and may be pre-molded for both the fully molded compensation ring and the drilled variant.

[0079] As stated above, the compensation ring 450 should be installed such that the tap ratio output is set to the first set point prior to locking the ring. The deviation in deflection from the first set point to the second set point, and so on, may be consistent in devices of this type; however due to variations in manufacturing of the camshaft, camshaft housing, and mating surfaces it cannot be assumed that the deviation from the “no deflection point” of the first set point is consistent between manufactured devices.

[0080] For the purposes of describing and defining the present invention it is noted that the terms “approximately” and “substantially” are utilized herein to represent the inherent degree of uncertainty that may be attributed to any quantitative comparison, value, measurement, or other representation. The terms “approximately” and “substantially” are also utilized herein to represent the degree by which a quantitative representation may vary from a stated reference without resulting in a change in the basic function of the subject matter at issue.

[0081] It is noted that recitations herein of a component of the present invention being “configured” in a particular way, “configured” to embody a particular property, or function in a particular manner, are structural recitations as opposed to recitations of intended use. More specifically, the references herein to the manner in which a component is “configured” denotes an existing physical condition of the component and, as such, is to be taken as a definite recitation of the structural characteristics of the component.

[0082] It is noted that one or more of the following claims utilize the term “wherein” as a transitional phrase. For the purposes of defining the present invention, it is noted that this term is introduced in the claims as an open-ended transitional phrase that is used to introduce a recitation of a series of characteristics of the structure and should be interpreted in like manner as the more commonly used open-ended preamble term “comprising.”

[0083] Although the disclosure has been illustrated and described herein with reference to explanatory embodiments and specific examples thereof, it will be readily apparent to those of ordinary skill in the art that other embodiments and examples can perform similar functions and/or achieve like results. For instance, the connection port insert may be configured as individual sleeves that are inserted into a passageway of a device, thereby allowing the selection of different configurations of connector ports for a device to tailor the device to the desired external connector. All such equivalent embodiments and examples are within the spirit and scope of the disclosure and are intended to be covered by the appended claims. It will also be apparent to those skilled in the art that various modifications and variations can be made to the concepts disclosed without departing from the spirit and scope of the same. Thus, it is intended that the present application cover the modifications and variations provided they come within the scope of the appended claims and their equivalents.

Claims

1. A fiber optic terminal comprising: a shell comprising a cavity; a plurality of ports comprising a control port having a port opening extending from an outer surface of the terminal into the cavity and defining a port passageway along a longitudinal axis; and a variable ratio coupler disposed within the cavity, the variable ratio coupler comprising a housing, an optical input, a first optical output, a second optical output, a shaft within the control port, a detent radially extending from the shaft, and a compensation ring disposed around the shaft, wherein: the compensation ring comprises a ring body, at least one arcuate slot within the ring body operable to receive at least one locking fastener, and a shaft opening within the ring body; the compensation ring is secured to the

housing by one or more locking fasteners; the at least one arcuate slot is at least partially surrounds the shaft opening; the shaft opening defines an inner surface; the inner surface comprises a plurality of detent receiving features; rotation of the shaft changes an output power level between the first optical output and the second optical output at a coupling region; and the detent is operable to be positioned in a selected detent feature of the plurality of detent features.

2. The fiber optic terminal of claim 1, wherein the plurality of detent receiving features corresponds to a plurality of split ratio set points of the variable ratio fiber optic coupler.

3. The fiber optic terminal of claim 1, wherein the plurality of detent receiving features comprises a plurality of blind bores.

4. The fiber optic terminal of claim 1, wherein the plurality of detent receiving features comprises a plurality of through holes.

5. The fiber optic terminal of claim 1, wherein the at least one arcuate slot comprises a pair of arcuate slots.

6. The fiber optic terminal of claim 1, wherein the arcuate slots of the pair of arcuate slots oppose one another.

7. The fiber optic terminal of claim 1, wherein the plurality of ports further comprises: at least one input connection port; and a pass-through output connection port.

8. The fiber optic terminal of claim 1, wherein the plurality of detent receiving features correspond to a plurality of set points for the output power level between the first optical output and the second optical output.

9. The fiber optic terminal of claim 1, wherein: the variable ratio coupler further comprises: a cam positioned at a portion of the shaft; and a flexure comprising a first end that contacts a portion of the cam and a second end that is held immobile in the variable ratio coupler; and a coupler, wherein each end of the coupler is mounted to a different end of the flexure; and rotation of the shaft causes the cam to change an amount of bend on the coupling region thereby changing the output power level between the first optical output and the second optical output at the coupling region.

10. The fiber optic terminal of claim 9, wherein rotation of the cam changes an amount of deflection on the first end of the flexure which also changes the amount of bend on the coupling region.

11. The fiber optic terminal of claim 1, wherein the plurality of ports are provided in an array at an end face of the shell.

12. A method of changing a split ratio of a fiber optic terminal, the method comprising: inserting an axle of a tool into a control port passageway of a control port of a plurality of ports of the fiber optic terminal to engage an end of the axle with a shaft of a variable ratio coupler within a cavity of the fiber optic terminal; and rotating the axle of the tool to rotate the shaft of the variable ratio coupler to a desired split ratio set point of a plurality of split ratio set points, wherein the desired split ratio set point corresponds to a desired output power level between a first optical output and a second optical output of the variable ratio coupler, and each split ratio set point is defined by a detent receiving feature of a plurality of detent receiving features of a compensation ring surrounding the shaft.

13. The method of claim 12, wherein the first optical output is provided at a first output connection port of the plurality of ports and the second optical output is provided at a second output connection port of the plurality of ports.

14. The method of claim 12, wherein the plurality of ports extend from an outer surface of a shell of the fiber optic terminal into the cavity defined by the shell.

15. The method of claim 12, wherein: the compensation ring comprises a ring body, at least one arcuate slot within the ring body operable to receive at least one locking fastener, and a shaft opening within the ring body; the compensation ring is secured to a housing by one or more fasteners; the at least one arcuate slot is at least partially surrounds the shaft opening; the shaft opening defines an inner surface; and the plurality of detent receiving features are positioned at the

inner surface.

16. The method of claim 15, wherein the plurality of detent receiving features corresponds to the plurality of split ratio set points of the variable ratio fiber optic coupler.

17. The method of claim 15, wherein the plurality of detent receiving features comprises a plurality of blind bores.

18. The method of claim 15, wherein the plurality of detent receiving features comprises a plurality of through holes.

19. The method of claim 15, wherein the at least one arcuate slot comprises a pair of arcuate slots.

20. The method of claim 19, wherein each arcuate slot of the pair of arcuate slots oppose one another.

21. The method of claim 12, wherein the plurality of ports further comprises: at least one input connection port; and a pass-through output connection port.

22. A variable ratio coupler for changing a split ratio of optical power for two optical fibers comprising: a coupler comprising a first optical output, a second optical output, and a coupling region operable to change the split ratio of the optical power for the two optical fibers at the first optical output and the second optical output; a housing; a shaft disposed within the housing and comprising an end, a radially extending detent, an end face at the end configured to receive a tool, wherein rotation of the shaft by the tool changes an output power level between the first optical output and the second optical output at the coupling region; a compensation ring disposed around the shaft, the compensation ring comprising: a ring body; at least one arcuate slot within the ring body; at least one locking fastener disposed within the at least one arcuate slot that secures the compensation ring to the housing; a shaft opening within the ring body, wherein: the at least one arcuate slot is at least partially surrounds the shaft opening; the shaft opening defines an inner surface; and the inner surface comprises a plurality of detent receiving features operable to receive the detent of the shaft; and a cam disposed on the shaft, wherein the cam has a shape that changes an amount of bend on the coupling region for a desired split ratio of optical power for the two optical fibers, and the plurality of detent receiving features corresponds to a plurality of split ratios of optical power.

23. The variable ratio coupler of claim 22, wherein the plurality of detent receiving features comprises a plurality of blind bores.

24. The variable ratio coupler of claim 22, wherein the plurality of detent receiving features comprises a plurality of through holes.

25. The variable ratio coupler of claim 22, wherein the at least one arcuate slot comprises a pair of arcuate slots.

26. The variable ratio coupler of claim 25, wherein each arcuate slot of the pair of arcuate slots oppose one another.

27. The variable ratio coupler of claim 22, wherein the cam comprises a plurality of zones, wherein a radius changes uniformly in each zone of the plurality of zones, and a rate of change of the radius is different among individual zones of the plurality of zones.

28. The variable ratio coupler of claim 27, wherein: the plurality of zones comprises a first zone, a second zone, and a third zone; in the first zone, the split ratio changes from a first split ratio to a second split ratio; in the second zone, the split ratio changes from the second split ratio to a third split ratio; and in the third zone, the split ratio changes from the third split ratio to the first split ratio.

29. The variable ratio coupler of claim 27, wherein: the plurality of zones comprises a first zone and a second zone; and the first zone has a range of rotation that is less than a range of rotation of the second zone.

30. A method of assembling a variable ratio coupler comprising: positioning a compensation ring over a shaft such that a detent is positioned within a detent receiving feature corresponding to a first split ratio set point; inputting an optical signal into an optical input of the variable ratio

coupler; measuring an optical power at a first optical output and a second optical output; turning the shaft until a split ratio of optical power between the first optical output and the second optical output is at the first split ratio set point; and securing the compensation ring to a housing of the variable ratio coupler.

31. The method of claim 30, wherein the compensation ring further comprises: a ring body; at least one arcuate slot within the ring body operable to receive at least one locking fastener; a shaft opening within the ring body, wherein: the at least one arcuate slot is at least partially surrounds the shaft opening; the shaft opening defines an inner surface; and the inner surface comprises a plurality of detent receiving features such that the detent receiving feature corresponding to the first split ratio set point is an individual one of the plurality of detent receiving features.

32. The method of claim 31, wherein securing the compensation ring to the housing comprises tightening the at least one locking fastener into the housing.

33. The method of claim 30, wherein the plurality of detent receiving features comprises a plurality of blind bores.

34. The method of claim 30, wherein the plurality of detent receiving features comprises a plurality of through holes.

35. The method of claim 30, wherein the at least one arcuate slot comprises a pair of arcuate slots.

36. The method of claim 35, wherein each arcuate slot of the pair of arcuate slots oppose one another.

37. A method of assembling a variable ratio coupler comprising: positioning a compensation ring over a shaft such that a detent is positioned within a detent receiving feature corresponding to a first split ratio set point; inputting an optical signal into an optical input of the variable ratio coupler; measuring an optical power at a first optical output and a second optical output; turning the shaft until a split ratio of optical power between the first optical output and the second optical output is at the first split ratio optical power set point; and recording the position of the shaft; sequentially turning the shaft until the split ratio of optical power is at a plurality of split ratio set points and recording the position of the shaft at each individual split ratio set point of the plurality of split ratio set points; removing the compensation ring from the shaft; fabricating a plurality of detent receiving features on the compensation ring according to the recorded positions of the shaft at the plurality of split ratio set points; positioning the compensation ring over the shaft such that the detent is positioned within the detent receiving feature corresponding to the first split ratio set point; inputting the optical signal into the optical input of the variable ratio coupler; measuring the optical power at the first optical output and the second optical output; turning the shaft until the split ratio of optical power between the first optical output and the second optical output is at the first split ratio optical power set point; and securing the compensation ring to a housing of the variable ratio coupler.

38. The method of claim 37, wherein the compensation ring further comprises a ring body and at least one arcuate slot within the ring body operable to receive at least one locking fastener.

39. The method of claim 38, wherein securing the compensation ring to the housing comprises tightening the at least one locking fastener into the housing.

40. The method of claim 37, wherein the plurality of detent receiving features comprises a plurality of blind bores.

41. The method of claim 37, wherein the plurality of detent receiving features comprises a plurality of through holes.

42. The method of claim 37, wherein the at least one arcuate slot comprises a pair of arcuate slots.

43. The method of claim 42, wherein each arcuate slot of the pair of arcuate slots oppose one another.

44. A compensation ring for a variable ratio fiber optic coupler, the compensation ring comprising: a ring body; at least one arcuate slot within the ring body operable to receive at least one locking fastener; a shaft opening within the ring body, wherein: the at least one arcuate slot is at least

partially surrounds the shaft opening; the shaft opening defines an inner surface; and the inner surface comprises a plurality of detent receiving features.

45. The compensation ring of claim 44, wherein the plurality of detent receiving features corresponds to a plurality of split ratio set points of the variable ratio fiber optic coupler.

46. The compensation ring of claim 44, wherein the plurality of detent receiving features comprises a plurality of blind bores.

47. The compensation ring of claim 44, wherein the plurality of detent receiving features comprises a plurality of through holes.

48. The compensation ring of claim 44, wherein the at least one arcuate slot comprises a pair of arcuate slots.

49. The compensation ring of claim 48, wherein each arcuate slot of the pair of arcuate slots oppose one another.
